

Jason Zheng

215-833-5883 | jzheng2413@gmail.com | <http://linkedin.com/in/jason-zheng01> | <http://github.com/Jason-Z1> <https://portfolio-jasonzheng.vercel.app/>

PROFESSIONAL SUMMARY

Full-stack, Embedded systems, and AI software engineer with client-facing consulting experience gathering requirements, synthesizing data-driven insights, and delivering iterative solutions under Agile methodologies. Comfortable moving from stakeholder interviews to database design to production systems, with a track record of translating ambiguous client needs into working software and clear documentation.

EDUCATION

Rensselaer Polytechnic Institute

Expected May 2028

B.S. Computer Science & Information Technologies and Web Science

GPA: 3.68/4.0

- Relevant Coursework: Computer Architecture & OS, Web Systems Development, Data Structures & Algorithms, Principles of Software

EXPERIENCE

NYSTEC – IgniteU Fellow

August 2026 – Present

- Conducted structured stakeholder interviews with 4 client stakeholders to define project scope and success criteria for a Custom GPT engagement, translating open-ended business goals into a concrete technical requirements document.
- Collected and synthesized user usage data across 6 use cases to identify the highest-value workflows, prioritizing which capabilities to build first based on client impact and feasibility.
- Applied Agile methodologies to iteratively build, test, and refine the Custom GPT, adapting the scope in response to evolving stakeholder feedback.

AgileRadioCom – Software Development Intern

May 2026 – August 2026

- Built a Python Flask + Socket.IO web server to orchestrate a 5+ device pipeline, routing real-time commands from browser input to embedded hardware over both UDP and RF transport layers.
- Designed REST endpoints to bridge web-layer commands to hardware-level actuation; contributed to a jam-detection ML model by handling data retrieval and cleaning, and implemented approximately 60-75% of the surrounding system functions, including adaptive switching of frequency, waveform, or modulation scheme in response to detected interference
- Engineered an event-driven full-duplex loop protocol for image retrieval over RF using an in-band countdown in status frames, backstopped by an ACK handshake to confirm command receipt before committing to a TX/RX flip.

PROJECTS

SyallbusAI

Jan. 2026 – May 2026

FastAPI, RAG (Groq + Gemini), Next.js, NextAuth, Docker

- Built a retrieval-augmented generation pipeline using Groq + Gemini APIs to parse and structure unformatted data into normalized, queryable tables via REST endpoints.
- Engineered a token-based rate limiting system with IP fallback to control API call volume, preventing abuse while maintaining UX for authenticated users.

Go Gaia

Sept. 2025

Microsoft Azure Computer Vision, Flask, Git/GitHub Collaboration

- Trained a Microsoft Azure Custom Vision model to classify object recyclability from images (~60% accuracy), then refactored from classification to detection, expanding the training set to ~600 cases and improving the accuracy.

Data Pulse

Sept. 2025 – Dec. 2025

Flask, Pandas, React, Bloomberg Terminal

- Designed and implemented a data routing pipeline to parse 10M+ rows of OHLCV stock data across 4,000+ equities, ingesting a decade of historical records from Bloomberg Terminal.
- Reduced average data lookup and filtration time from 20 minutes to 5 minutes (75% improvement)
- Built a full-stack dashboard with React + Recharts enabling both tabular and graphical visualization of filtered datasets via a Flask REST API backend.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R, MATLAB

Frameworks & Libraries: React, Node.js, Flask, JUnit, jQuery, FastAPI, Tailwind, PyArrow, Stable-Baselines3

Tools: Git, Docker, VS Code, Visual Studio, GenAI, Postman API, Bloomberg Terminal